

Shell & Tube Exchanger engineering manual

Compare heat duty with design and track overall heat transfer, fouling resistance and effectiveness.

Run the Python model

Open shell_tube_exchanger_engine.py in Python IDLE. Select Run, then Run Module (F5). Python 3.10 or newer is required; no extra packages or other files are needed.

The built-in example writes a results JSON file beside the Python file. Edit SETTINGS near the top and run again to change assumptions. To use your own measurements, run the file with an input JSON filename. Inputs use the fields and units below.

The input envelope contains schemaVersion, model, source, timezone, config and measurements. Each measurement contains timestamp, values and optional quality. Use ISO timestamps with an offset. Bad, stale and missing signals remain unavailable. The AVEVA mapping is provisional; a live connection has not been verified.

Inputs and units

Model inputs · 4 rows

INPUT	UNIT	USE
Timestamp; hot and cold inlet/outlet temperatures	date/time; °C	Four temperatures for duty and temperature driving force.
Hot and cold mass flow and heat capacity	kg/h; kJ/(kg·K)	Both stream duties and heat-capacity rates.
Shell and tube differential pressure	bar	Optional measured pressure drops; retained in results.
Shell passes, heat-transfer area, clean U and design duty	count; m²; W/(m²·K); MW	Configuration for correction factor and reference comparisons.

Calculations

DUTY

$$Q_{\text{hot}} = \dot{m}_{\text{hot}} C_{p,\text{hot}} (T_{\text{h,in}} - T_{\text{h,out}}) / 3600; Q_{\text{cold}} = \dot{m}_{\text{cold}} C_{p,\text{cold}} (T_{\text{c,out}} - T_{\text{c,in}}) / 3600$$

kW. Q_{avg} is their mean; imbalance = $100 \times |Q_{\text{hot}} - Q_{\text{cold}}| / |Q_{\text{avg}}|$.

TEMPERATURE DRIVING FORCE

$$\text{LMTD} = (\Delta T_1 - \Delta T_2) / \ln(\Delta T_1 / \Delta T_2)$$

$\Delta T_1 = T_{\text{h,in}} - T_{\text{c,out}}$; $\Delta T_2 = T_{\text{h,out}} - T_{\text{c,in}}$. Equal end differences use that common difference.

PASS CORRECTION

$$P = (T_{c,out} - T_{c,in}) / (T_{h,in} - T_{c,in}); R = (T_{h,in} - T_{h,out}) / (T_{c,out} - T_{c,in})$$

The N-shell-pass relation supplies $F(P,R,N)$. Effective driving force = $F \times \text{LMTD}$.

HEAT TRANSFER AND FOULING

$$U = 1000 \times Q_{avg} / (A \times F \times \text{LMTD}); R_f = 1/U - 1/U_{clean}$$

R_f is displayed in $10^{-4} \text{ m}^2 \cdot \text{K}/\text{W}$. Negative values mean performance exceeds the chosen clean reference.

EFFECTIVENESS AND DESIGN COMPARISON

$$\epsilon = 100 \times Q_{avg} / [C_{min}(T_{h,in} - T_{c,in})]$$

$C = \dot{m}C_p/3600$ in kW/K. Duty deviation = $100 \times (Q_{avg,MW} - Q_{design})/Q_{design}$.

Useful trends

- Average duty versus design and hot/cold duties to check balance.
- Overall U and fouling resistance at comparable flow rates.
- Effectiveness, LMTD, correction factor and both end approaches.

Assumptions and limits

- Single-phase sensible heat with constant C_p . The implemented arrangement is N shell passes with the corresponding even-tube-pass correction, not arbitrary exchanger geometry.
- Separate shell-side and tube-side film coefficients are not calculated. They require geometry, fluid properties and suitable correlations.
- Invalid terminal temperatures or an undefined correction factor suppress U and fouling results. Low U alone does not establish fouling; first check flows, temperatures and balance.

Measured values come from instruments or samples. Calculated values use those inputs and the configured assumptions. Design values are references. Simulated values demonstrate the model and are not plant measurements. Missing results remain unavailable.

Sources

- Supplied Heat Exchanger workbook: duty, effectiveness and pass arrangements
- Bowman, Mueller and Nagle (1940): Mean Temperature Difference in Design, N-shell-pass correction